SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Total Marks: 25

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Code of Conduct

I C.Divya Charan (192525113) **(Name / Reg. No.)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: C.Divya Charan

Assessment Tasks
Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. **Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.**

1) Definition :-

Cloud computing mode in which resources such as servers, storage, databases, networking, software and processing power are provided to users through the internet on demand instead of purchasing and maintaining physical infrastructure, organizations can obtain computing resource from cloud service provider according to their requirement.

Characteristics :-

1. On-Demand Self Service :-

Users can independently use computing resources wherever they are required.

2. Broad Network Access :-

Cloud services can be accessed through the internet using different devices such as computing, laptop tablets, and smartphones. This allows user to access applications and data from different Locations.

3. Resource Pooling :-

Cloud providers maintain a shared pool of computing resources such as CPU, memory etc. These resources are dynamically allocated to multi users according to their requirements.

4. Rapid Elasticity :-

Cloud Resources can be quickly increases or decreases according to workload

Cloud Computing	On-Premise Computing
* Resources are accessed through the Internet	* Resources are maintained by the organization.
* Lower Initial Infrastructure investment	* High initial infrastructure investment
* Resources can be scaled quickly	* Scaling requires additional hardware.

Conclusion :-

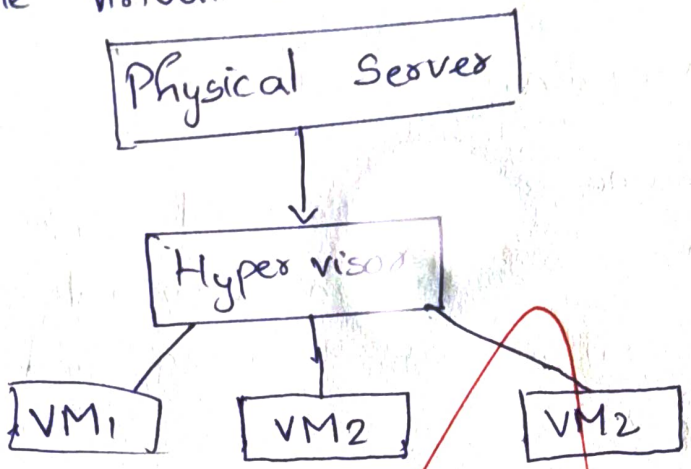
Cloud computing provides flexibility, scalability, accessibility and efficient resource utilization. its characteristics make it more adaptable than traditional-on-premise computing and suitable for modern application & organization.

4. Distributed Computing :-

It allows multiple computing to work together to perform tasks.

5. Virtualizations :-

Virtualization include one physical server to run multiple virtual machines.



Conclusion :-

Cloud computing evolved gradually from hardware and networking technologies to internet-based software services.

Each stage improved resourced sharing, connectivity and accessibility.

ntroduction:-

Servers virtualization is a fundamental technology used in cloud computing. In traditional computing, one physical servers may be dedicated to one application, which can result in virtualization

Server Virtualization:-

It is the virtual Machine. Each machi can run it's own OS.

It's Role in Cloud Computing:-

Resource Utilization:-

It allows multiple virtual machines to sh resources of one physical server

Cost Reduction:-

It reduces the cost needed for physical hardware and also lowers the maintenan cost.

Scalability:-

Virtual Machines can be created or removed according to user requirements.

Flexibility:-

Different Operating Systems and applications run on the same physical servers.

Resource Management:-

CPU, memory and storage can be allocated

Disaster Recovery:-

Virtual Machines can be backed up & restored when required.

Difference Between virtualization & cloud computing

Virtualization	cloud Computing
* It is a enabling technology	* It is a service giving model
* It creates virtual versions of physical resources	* It provides computing resources over the internet.
* It can operate with in a single organization	* It provides services to web on demand

Conclusion:-

Server Virtualization is an important technology that improves resource utilization and reduce costs. Cloud Computing uses virtualization and other technologies to deliver flexible, scalable & no-demand services

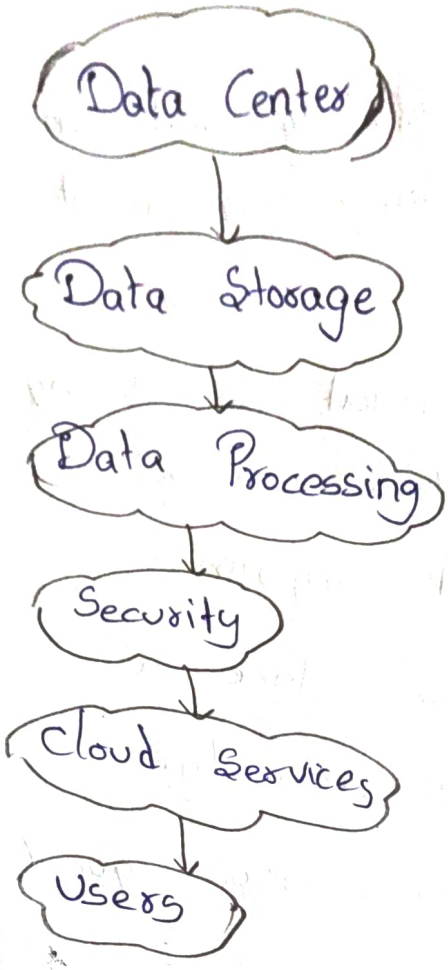
Introduction:-

A data center is a facility that contains servers, storage devices, networking equipment and supporting infrastructure.

Data center are essential for storing, processing and managing and delivering cloud services.

Role of Data Centers in Cloud Computing:-

1. Data Storage:- Data Centers store files, databases application Data, and backups securely.
2. Data Processing:- Servers Process uses request and run cloud application.
3. Network Connectivity:- Networking Equipment connects Servers to internet and enables users to access cloud services.
4. High Availability:- Backup power supplies, redundant equipment and backup systems help reduce service Interruption
5. Scalability:- Data Centers provides additional Computing resources when user demand increases.
6. Disaster Recovery:- Backup systems and geographic Seperate facilities help restore service after failures



Conclusion:-

Data centers form the physical foundation of cloud computing. They ensure that the cloud services are available secure, scalable and reliable for users world wide

troduction :-

Cloud Computing provides different service models based on the needs on the needs of users and organization. The major models are Infrastructure as aservice (IaaS), Platform as a Service (PaaS), Software as a Service (SaaS) and anything as a Service and Anything as a Service (XaaS)

Infrastructure as a Service (IaaS)

IaaS provides virtual computing resourcing such as servers, storage and networking over the Internet. Users can install and manage their operating system and application.

Platform as a Service (PaaS)

PaaS provide a platform and development tools for building, testing and deploying applications without managing the underlying hardware.

Software as a Service (SaaS)

SaaS provides ready to use software o Users can access applications through a web browse without installing or maintaining the underlying infr

4. Anything as a Service (

XaaS is a broad term that refers to delivering different types of IT resources and services ther internet. On demand. It can include Infrastructure, Software, Platforms, Storage & security Services.

Comparison of Cloud Service Models

Feature	IaaS	PaaS	SaaS	XaaS
Full Form	Infrastructure as a Service	Platform as a service	Software as a service	anything as a service.
main purpose	Provides Infrastructure	Provides development Platforms	Provides Ready	Provides various Services.

Conclusion:-

IaaS provides infrastructure, PaaS provides application development Platforms and SaaS provide ready-to-User software

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